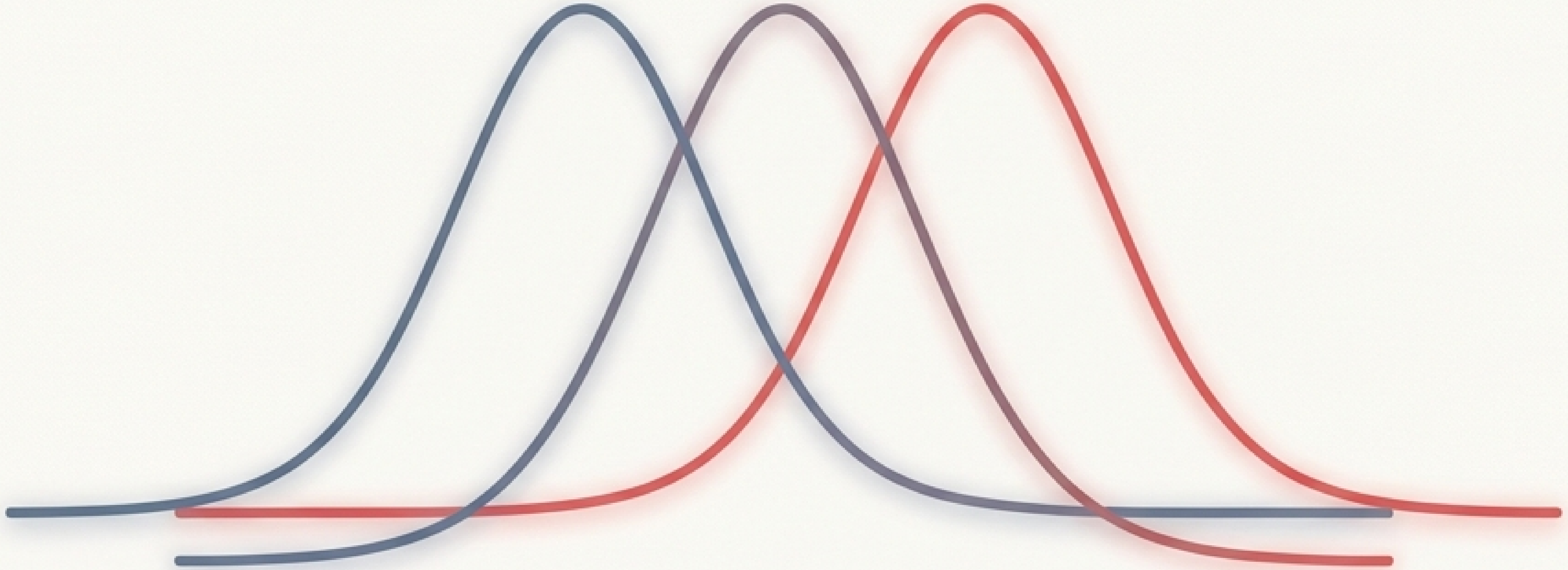


Distinguishing Signal from Noise: The ANOVA Test

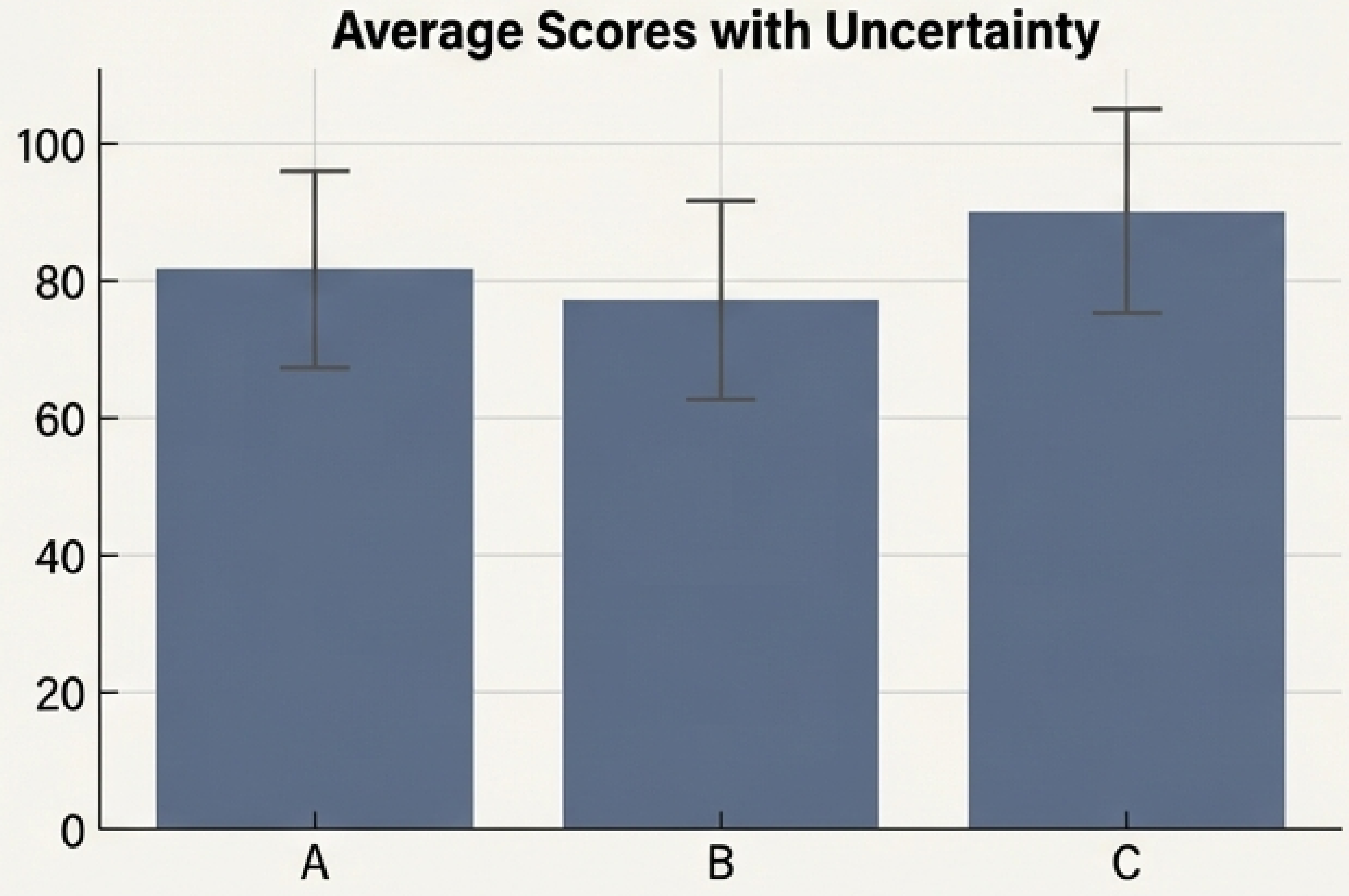
A guide to implementing and understanding Analysis of Variance for continuous data.

Designed for self-paced study. This deck bridges the gap between statistical theory and Python implementation.

THE PROBLEM

Why simply comparing averages is dangerous.

Suppose we test three teaching methods. The average scores look different: Method A (81.6), Method B (77.3), and Method C (90.0). Is Method C actually better, or did it just get lucky with smarter students?



THE SOLUTION & HYPOTHESIS

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The "Eye Test" fails in statistics. We need a rigorous way to determine if these differences are statistically significant or just random noise.

ANOVA (Analysis of Variance) compares the means of three or more groups to determine if at least one group mean is fundamentally different from the others.

Null Hypothesis (H_0):

All group means are equal (Status Quo).

$$H_0 : \mu_A = \mu_B = \mu_C$$

Alternate Hypothesis (H_1):

At **least one** group mean is **different** (Discovery).

$$H_1 : \text{Not all } \mu_i \text{ are equal}$$

NotebookLM

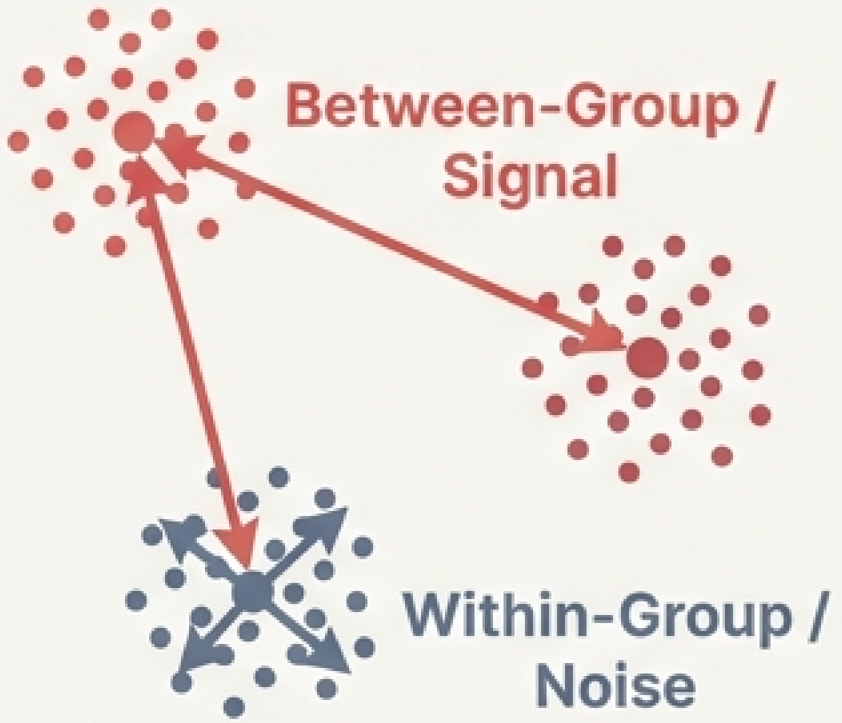
The Logic: Comparing the Signal against the Noise

$$F = \frac{\text{Variance between groups}}{\text{Variance within groups}}$$

(The Signal: Variance caused by the factor we are testing)

(The Noise: Random variance from individual differences)

High F-Value = Signal dominates Noise = Significant Result



The Mechanism: A six-step process to calculate certainty.

1. Calculate Means →
2. Calculate SSB (Signal) →
3. Calculate SSW (Noise) →
4. Degrees of Freedom →
5. Mean Squares →
6. F-Ratio

Reference Dataset

Group A	5	6	8
Group B	10	12	9
Group C	13	9	12

We will use this dataset to manually track the calculation through the next four slides.

Steps 1 & 2: Quantifying the differences between groups.

Step 1: The Means

$\bar{x}_A$ = 6.33

$\bar{x}_B$ = 10.33

$\bar{x}_C$ = 11.33

Overall Mean $(\bar{X})$ = 9.33

Step 2: Between-Group Sum of Squares (SSB)

$$SSB = \sum n_i(\bar{x}_i - \bar{X})^2$$

$3 \times \left[(6.33 - 9.33)^2 + (10.33 - 9.33)^2 + (11.33 - 9.33)^2\right]$

SSB = 42

The Signal (Numerator)

Step 3: Quantifying the internal variance (The Noise).

We measure how much data points vary within their own groups, igroring group differences.

$$SSW = \sum\sum(X_{ij} - \bar{x}_i)^2$$

Group A:	$(5-6.33)^2 + (6-6.33)^2 + (8-6.33)^2 = 4.65$
Group B:	$(10-10.33)^2 + (12-10.33)^2 + (9-10.33)^2 = 4.65$
Group C:	$(13-11.33)^2 + (9-11.33)^2 + (12-11.33)^2 = 8.65$

Total SSW = 4.65 + 4.65 + 8.65 = **17.95**

The Noise (Denominator)

Steps 4 & 5: Normalising the variance.

Band 1: Degrees of Freedom (df)

We adjust for sample size.

$df_{\text{between}} = k - 1$ (3 groups - 1) $\rightarrow$ **2**

$df_{\text{within}} = N - k$ (9 samples - 3 groups) $\rightarrow$ **6**

Band 2: Mean Squares (MS)

Divide the Sum of Squares by Degrees of Freedom to get average variance.

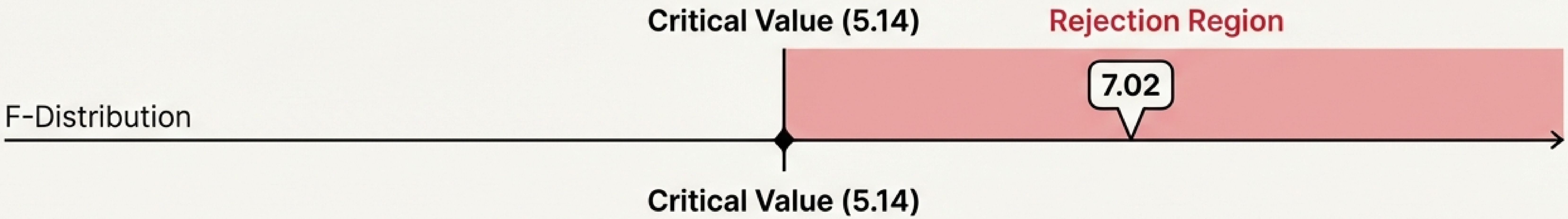
MSB (Mean Square Between) $= \frac{42}{2} = $ **21**

MSW (Mean Square Within) = $\frac{17.95}{6} \approx$ **2.99**

Step 6: The Verdict

F = 7.02

$$F = \frac{MSB}{MSW} = \frac{21}{2.99}$$



Since 7.02 > 5.14, the difference is significant.

Result: Reject H_0. There is a statistically significant difference between the groups.

Implementation: Automating ANOVA with Python.

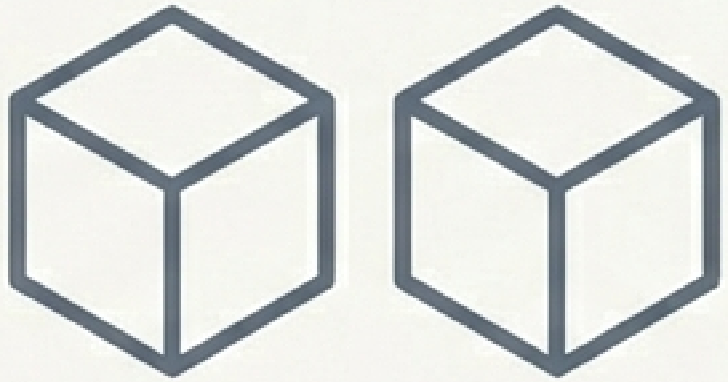
```python
from scipy import stats

# Define the groups
group_a = [5, 6, 8]
group_b = [10, 12, 9]
group_c = [13, 9, 12]

# Run the One-Way ANOVA
f_stat, p_value = stats.f_oneway(group_a, group_b, group_c)

print(f"F-statistic: {f_stat}")
print(f"p-value: {p_value}")
```

Terminal Output

```
F-statistic: 7.02
p-value: 0.0268
```

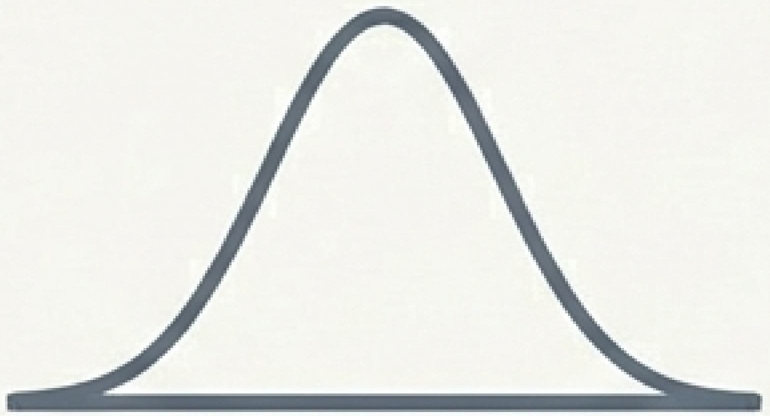
p-value < 0.05
→ Reject Null Hypothesis.

Prerequisites: When is ANOVA valid?



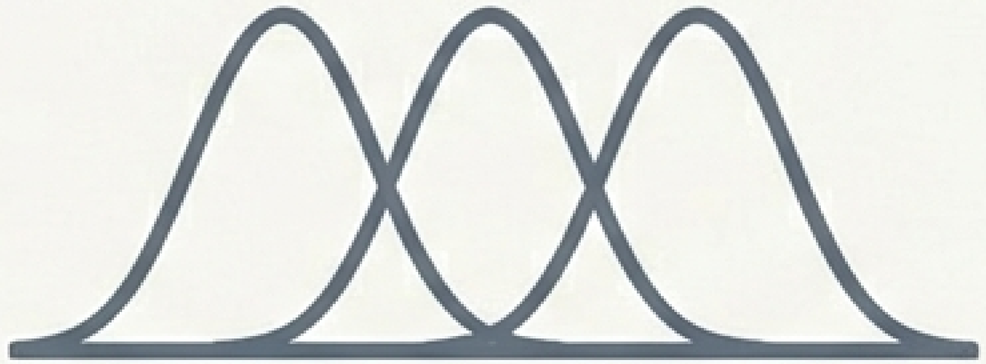
Independence

Samples must be independent of each other. One subject's result cannot influence another's.



Normality

The residuals (errors) within each group should follow a normal distribution.



Homogeneity of Variance

The variance (spread) within each group should be roughly equal across all groups.

⚠ Violating these assumptions can lead to **'False Positives'** (Type I errors).

Selecting the right test for the variable.

Test Type	Usage (Independent Variables)	Example Scenario
One-Way ANOVA	Used for **1 independent variable.**	e.g., Comparing yield of 3 different fertilisers.
Two-Way ANOVA	Used for **2 independent variables.**	e.g., Comparing fertilisers AND watering frequency.
Repeated Measures	Used for the **same subjects** across conditions.	e.g., Measuring the same patients' blood pressure at Week 1, 4, and 8.

Applied Scenarios: Education & Agriculture

Education

Exam Score Comparison

- Method A: 78, 85, 82
- Method B: 75, 80, 77
- Method C: 90, 88, 92

At 5% significance level, is there a significant difference in mean scores among the three teaching methods?

Agriculture

Crop Yield Optimisation

- Fertiliser A: 20, 22, 21 kg
- Fertiliser B: 25, 27, 26 kg
- Fertiliser C: 23, 24, 22 kg

Is there any statistically significant difference in the average yield among these fertilisers?

Applied Scenarios: Business & Technology

Card 3: Business

Customer Satisfaction

Testing 3 Customer Service Training Programmes

Program A: 3.5, 4.0, 3.8
Program B: 4.2, 4.5, 4.1
Program C: 3.9, 3.7, 4.0

Do mean satisfaction scores differ significantly among the three training programmes?

Card 4: Technology

Website Engagement Time

A/B/n testing of 3 homepage layouts

Layout A: 4.2, 4.5, 4.0 min
Layout B: 5.1, 5.0, 5.3 min
Layout C: 4.4, 4.1, 4.3 min

Is there a significant difference in time spent on the website across different homepage layouts?

Applied Scenarios: Clinical Trials

Blood Pressure Medication

A researcher compares blood pressure reduction across three medications.

Medication A:
8, 9, 7 mmHg

Medication B:
10, 11, 12 mmHg

Medication C:
9, 10, 8 mmHg

Does the average blood pressure reduction differ across the medications?

In this context, ANOVA helps identify the most effective drug candidate for further testing, moving beyond simple observation to statistical proof.

Summary: The Decision Framework.

1.

Purpose

ANOVA tests mean differences across 3+ groups.

2.

The Statistic

It uses the **F-ratio** to compare:
Between-Group Variance (Signal)
vs.
Within-Group Variance (Noise)

3.

The Trigger

A small p-value (< 0.05) indicates the groups are not all the same; the **Null Hypothesis** must be rejected.

ANOVA is the gateway to experimental design and inferential statistics, moving us from “it looks different” to “it *is* different”.